

# Dhruv Venkat

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## Education

### University of Waterloo

Bachelor of Applied Science, Computer Engineering (Honours)

Sep 2024 - 2029

Waterloo, ON

## Professional Experience

### Red Lane Group

Software Development Intern

May 2026 - Present

Calgary, AB

- Built a real-time **.NET recording service** capturing **100% of support calls**, replacing a legacy batch system
- **Cut per-call post-processing latency 20+ min** via a feature-flagged streaming-SDK transcription pipeline
- Productionized an **event-driven Zendesk triage pipeline deployed using serverless functions**, projecting **100+ hrs saved/month**

### Scotiabank

Software Engineering Intern

Sep 2025 - Dec 2025

Toronto, ON

- Built **RAG agents** (LangChain/FAISS) to generate technical requirements **40% faster**, **now in production**
- Architected a **Tesseract OCR pipeline** for scanned documents, **tripling** institutional processing throughput
- Migrated **8 enterprise applications** from Java 11 to Java 21 (LTS), **cutting projected maintenance by 50%**

### Tangerine

DevOps Engineering Intern

Jan 2025 - Apr 2025

Toronto, ON

- Maintained **multi-branch Jenkins CI/CD pipelines** for development teams, **improving build reliability 30%**
- Automated **Python security scans** across **100+ GitHub repositories**, cutting severe vulnerabilities by 20%
- Built a GitHub Copilot usage tracker (REST APIs, Apache Groovy); **in production, saving \$1000+/month**

## Projects

### Firmware Developer (UW Orbital) | Embedded C, FreeRTOS, I2C, SPI, Python [URL](#)

- Developed a diagnostic command to **downlink register configurations** for the CC1120 transceiver **over SPI**
- Wrote a register-level **I2C driver for the LM75BD sensor** with **threshold-based thermal response**

### DynaEQ: Adaptive Linux Audio Daemon | C, Linux, MPRIIS, EasyEffects, Make [URL](#)

- Built a modular **C daemon** that adapts system EQ to live playback monitored over **MPRIIS** in real time
- Designed **fixed-size EQ profiles** and static lookup tables for **bounded-memory**, deterministic control
- Added **safe-default fallbacks**, cleanup paths, and suppression of redundant EQ backend writes

### flowstate: Terminal-Native C++ Code Editor | C++, POSIX TTY, CMake, Git [URL](#)

- Built a **terminal-native C++ editor** with **POSIX termios raw-mode I/O** and a custom renderer
- Added **syntax highlighting and LSP diagnostics** for C/C++, Python, TypeScript, Go, and Rust
- Integrated a **Codex-powered AI assistant** for contextual explanations, code generation, and in-place edits

### linkdrop: LAN-Based Link Sharing Daemon | Go, Linux, BASH, SQLite, HTTP [URL](#)

- Built a **Go daemon** that shares links from mobile to Linux PCs over **LAN**, no cloud sync
- Implemented **HTTP endpoints** and a lightweight browser UI to receive, store, and open links in one action
- Wrote **Linux install scripts** that provision local app data and register the daemon as a **systemd service**

## Technical Proficiencies

**Languages:** C, Embedded C, C++, RISC-V Assembly, Python, Go, TypeScript, Java, SQL

**Embedded & Systems:** FreeRTOS, I2C, device drivers, POSIX/termios, Linux, CMake

**Tools & Cloud:** Git, Jenkins, AWS, Microsoft Azure, REST APIs, PostgreSQL, Node.js, React/Next.js